Smart Grids, Demand Response, and Distributed Energy Resources

SYNTHETIC EDUCATIONAL DOCUMENT — NOT AN OFFICIAL SOURCE

Purpose: Provide background material for retrieval and chatbot testing in the SmartGrid AI prototype.

This synthetic overview explains technologies used in modern electricity systems. It is intended for general education, not as a description of a specific utility deployment.

1. Smart grid

A smart grid uses digital monitoring, communications, automation, and data analytics to improve visibility and coordination across electricity networks. Examples include smart meters, automated switches, sensors, advanced control systems, and outage management software.

2. Smart meters

Smart meters can record electricity use at intervals and communicate readings to utilities. Data availability, interval length, privacy protections, and customer access vary by program and jurisdiction.

3. Demand response

Demand response changes electricity consumption in response to price signals, incentives, or reliability requests. Examples include adjusting thermostats, shifting industrial processes, or temporarily reducing nonessential loads. Programs must communicate eligibility, timing, and customer impacts clearly.

4. Distributed energy resources

Distributed resources include rooftop solar, batteries, controllable loads, and some small generators. Aggregating these resources can provide flexibility, but performance depends on interconnection rules, communications, customer participation, and operating conditions.

5. Battery storage

Batteries can charge when electricity is available and discharge later. Their contribution depends on power rating, energy capacity, state of charge, duration, efficiency, and dispatch strategy. A battery's rated capacity does not mean it can supply that output indefinitely.

6. Microgrids

A microgrid is a group of loads and distributed resources that can operate in coordination and, when designed for it, disconnect from the larger grid. Not every facility with solar panels has islanding capability.

7. Cybersecurity and privacy

Connected grid devices create cybersecurity and data-protection requirements. Systems should use access controls, secure communications, monitoring, patch management, and privacy safeguards appropriate to their risk.

End of synthetic document. Verify operational facts with current official sources before real-world use.